Shuwen Wang

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Education

- **Northeastern University** Boston, MA
  Master of Science in Computer Science; GPA: 3.9 *Sept. 2023 – Jan. 2025*
  Courses: Machine Learning, Web Programming, Computer Network, Computer System, Operation System, Natural Language Programming

- **University Of Washington** Seattle, WA
  B.S. in Applied Mathematics; GPA: 3.6 *Sept. 2015 – Aug. 2019*

Programming skills

- **Languages:**: Python, Java, JavaScript (TypeScript), C/C++, SQL, Shell Script
- **Frameworks:**: Spring, React, Angular, Django, Flask, TensorFlow, PyTorch, Scikit-learn
- **AI/ML:**: LLMs, RAG, Multi-Agent Systems (LangGraph, ReAct), LoRA Fine-tuning, NLP, Deep Learning, Model Evaluation
- **Databases/Tools:**: DynamoDB, MongoDB, MySQL, Elasticsearch, Docker, Kubernetes, AWS (EC2, Lambda), Git

Experience

- **LexisNexis** June. 2024 – Present
  Machine Learning Engineer *Raleigh, NC*
  - Built and optimized LLM-powered multi-agent systems for complex workflow automation and analytical tasks
  - Designed RAG pipelines for high-quality information retrieval from structured and unstructured data sources
  - Developed orchestration strategies to route tasks across models, improving reasoning accuracy and system efficiency
  - Analyzed system performance and user behavior to improve output quality and reliability

- **Biogen** Jan. 2024 – June. 2024
  Software Engineer *Cambridge, MA*
  - Developed a modern open-source molecular alignment and 3D similarity scoring algorithm, named Roshambo. Utilizing C++ and CUDA for parallel processing. The algorithm efficiently operates across 1 GPU and 24 CPUs, producing aligned molecules with both shape and color similarity
  - Trained and deployed **CNN**, **RNN** and **LSTM** modules on AWS using PyTorch, Tensorflow and other Deep Learning frameworks to predict drug–drug interactions
  - Contributed to the growth of Biogen's Molecular Informatics open-source project, utilizing Flask, React, and Kubernetes to create a scalable and maintainable architecture. Managed the project's infrastructure on AWS, ensuring seamless deployment and updates.

- **Growth Channel** Mar. 2022 – Aug. 2022
  Software Engineer *Austin, TX*
  - Implemented a Dashboard module in the front end using **React**, based on **Figma** wireframes, including an activity stream, a website KPI growth graph, and visualizations of direct marketing views and strategies for over 2,000 users.
  - Maintained and developed **AWS Lambda** functions in the back end with **DynamoDB** and **MongoDB** using **Python**. Deployed a serverless application on **AWS Cloud** using **SAM CLI**.
  - Led a globally distributed **Scrum** team of 10 members across multiple countries, adopting **DevOps** practices.

Projects

- **LLM-powered Stock Investment Advisor**
  - Built a multi-agent (LangGraph + ReAct) system for end-to-end stock analysis (fundamentals, technicals, valuation, news)
  - Standardized financial data via MCP APIs, achieving ∼99% data consistency and 98% tool success rate
  - Fine-tuned a Qwen3-8B LoRA model on 100K financial news for sentiment/risk prediction (91% / 88% accuracy)
  - Reduced latency from 2+ min to ∼90s via parallelization and caching; reached ∼90% user satisfaction

- **Automated PPT Generation Agent**
  - Designed and implemented an automated PPT generation agent using LLMs, enabling end-to-end creation of presentation decks from user inputs and documents (PDF/PPT templates)
  - Architected a multi-stage agent pipeline (content understanding → outline planning → slide generation → template-based rendering) to ensure structured and high-quality outputs
  - Leveraged prompt engineering and function calling to produce structured slide content and integrate with programmatic PPT generation tools
  - Applied hierarchical clustering to organize unstructured input into coherent slide groupings, improving logical flow and presentation structure
  - Built an LLM-based evaluation (S-Judge) system to rank and select optimal outlines and slide content, enhancing consistency and output quality